MUSIC SALES ANALYSIS

One-line description: A denormalized Excel export combining 11 source tables (customers, invoices, invoice lines, tracks, albums, artists, genres, media types, playlists, playlist_track, employees) to support revenue, customer, product and operational analysis.

Dataset snapshot

- Rows: **11,561**
- Columns: **31**
- Date range (sales): **2017-01-03 → 2020-12-30**

Contents (primary tables represented)

- Customer (contact, company, support rep)
- Invoice / Invoice_line (invoice-level and line-level sales)
- Track → Album → Artist (product metadata: duration, composer)
- Genre, Media_type (taxonomies)
- Playlist / Playlist_Track (many-to-many track collections)
- Employee (support rep metadata)

Key data quality notes (high priority for readers)

- **Major join issue:** invoice lines are duplicated in the denormalized file due to joining Invoice_line with Playlist_Track (many-to-many). Aggregations on the flattened table will overcount revenue unless deduplicated by invoice_line_id.
- **Placeholders instead of NULLs:** many fields use 'Not available' (company etc.), inflating distinct counts and masking missing data.
- **Encoding corruption:** at least one customer record (repeated across ~186 rows) contains garbled characters in name/email — needs encoding/cleanup.
- **Monetary fields:** unit_price values are present but should be stored/treated as DECIMAL (10,2) to avoid float/rounding issues.

- **Dates:** invoice date exists as strings and was normalized to a DATE in cleaning; keep canonical DATE for analysis and use formatted strings only for presentation.
- **Track length outliers:** milliseconds contains extreme values (up to ~2.6M ms); verify long-duration tracks.

Core KPIs & analyses included / recommended

- Total revenue, Revenue by country / city
- Monthly revenue trend, Average Order Value (AOV)
- Top customers (LTV), repeat vs one-time customers
- Revenue by genre, top artists / tracks by revenue
- Invoice reconciliation: sum(invoice_line) ≈ invoice.total (validation included)
- Basket analysis: tracks frequently purchased together (prototype query provided)

Deliverables included for submission

- Cleaned canonical fact table: InvoiceLineFact (one row per invoice_line)
- Analytics-ready fact view: MUSIC_FACT (joins invoice lines → dimensions; excludes playlist multiplication)
- T-SQL cleaning & ETL script (full, ordered, idempotent) — ready to run in SSMS
- Sample business queries (top customers, monthly trend, product & genre performance)
- Recommendations / ETL checks (detection of duplicates, email validation, placeholder→NULL conversion)

OBJECTIVES:

- **Revenue Analysis:** Evaluate overall sales performance, revenue trends over time, and contribution by country, genre, artist, and track.
- **Customer Insights:** Identify high-value customers, purchasing patterns, repeat vs one-time buyers, and customer geographic distribution.
- **Product Performance:** Analyse which genres, albums, artists, and media types generate the most revenue to support catalogue and marketing decisions.
- **Operational Efficiency:** Assess support-rep performance, invoice patterns, and playlist impact on track popularity.
- **Data Quality Validation:** Ensure joined dataset accuracy by checking duplicates, missing values, and reconciliation of invoice totals vs line totals.

RESEARCH METHODOLOGIES:

1. Data Collection & Integration

- Combined 11 normalized tables (Customer, Invoice, Invoice_line, Track, Album, Artist, Genre, Media_type, Playlist, Playlist_Track, Employee) into a single analytics-ready dataset.
- Used primary-key / foreign-key joins (invoice_id, track_id, artist_id, etc.).

2. Data Cleaning & Validation

- Removed duplicate invoice-line records.
- Converted placeholders like *"Not available"* to NULL.
- Standardized date and currency formats.
- Verified revenue integrity:
  Sum(Invoice_line) ≈ Invoice.total

3. Exploratory Data Analysis (EDA)

- Used SQL, Excel, and Python to analyze:
  - Revenue by month, country, genre, artist.
  - Customer lifetime value.
  - Average order value.
  - Track popularity and playlist frequency.

4. Business Intelligence & Visualization

- Built dashboards in Tableau / Streamlit showing:
  - KPIs (Total Revenue, Customers, Orders)
  - Sales trends
  - Top customers and products
  - Geographic sales distribution.

5. Insight Generation

- Interpreted patterns to recommend:
  - Target markets
  - High-performing genres/artists
  - Customer retention strategies
  - Catalogue optimization.

ANALYSES

(A) Cleaning the tables

SQL - Clean ...hagata (77)) | SQL - Q&A.sq...thagata (86))

```sql
USE [PRACTICE_DB];
-- ============================
-- (A) Cleaning the track table
-- ============================
-- Set missing composer to default
UPDATE track
SET composer = 'Not available'
WHERE composer IS NULL;

-- Ensure unit_price has proper decimal format
ALTER TABLE track
ALTER COLUMN unit_price DECIMAL(10,2);

-- Verify cleaned track table
SELECT * FROM track;


-- ============================
-- (B) Cleaning the customer table
-- ============================
UPDATE customer

    -- Remove corrupted characters
SET first_name = ISNULL(LTRIM(RTRIM(REPLACE(REPLACE(REPLACE(first_name, '¿', ''), '½', ''), 'ï', ''))),'Not available'),
    last_name  = ISNULL(LTRIM(RTRIM(REPLACE(REPLACE(REPLACE(last_name, '¿', ''), '½', ''), 'ï', ''))),'Not available'),
    address    = ISNULL(LTRIM(RTRIM(REPLACE(REPLACE(REPLACE(address, '¿', ''), '½', ''), 'ï', ''))),'Not available'),
    city       = ISNULL(LTRIM(RTRIM(REPLACE(REPLACE(REPLACE(city, '¿', ''), '½', ''), 'ï', ''))),'Not available'),
    company    = ISNULL(LTRIM(RTRIM(REPLACE(REPLACE(REPLACE(company, '¿', ''), '½', ''), 'ï', ''))),'Not available'),
    state       = ISNULL(state, 'Not available'),
    fax         = ISNULL(fax, '0'),
    phone       = ISNULL(phone, '0'),
    postal_code = ISNULL(postal_code, '0');

-- Verify cleaned customer table
SELECT * FROM customer;
```

```sql
-- ================================
-- (C) Cleaning the employee table
-- ================================
-- Change reports_to to VARCHAR for storing titles
ALTER TABLE employee
ALTER COLUMN reports_to VARCHAR(50);

-- Update NULL or 0 reports_to
UPDATE employee
SET reports_to = 'Not available'
WHERE reports_to IS NULL OR reports_to = '0';

-- Replace numeric reports_to with manager's title
UPDATE e
SET e.reports_to = m.title
FROM employee e
INNER JOIN employee m
    ON e.reports_to = CAST(m.employee_id AS VARCHAR(50))
WHERE e.reports_to <> 'Not available';

-- Verify employee table
SELECT * FROM employee;
```

```sql
-- ==============================
-- (D) Cleaning the invoice table
-- ==============================
-- Step 1: Add new date-only column
ALTER TABLE invoice
ADD invoice_date_only DATE;

-- Step 2: Copy data into new column
UPDATE invoice
SET invoice_date_only = CAST(invoice_date AS DATE);

-- Step 3: Verify data
SELECT invoice_date, invoice_date_only
FROM invoice;

-- Step 4: Drop old column after verification
ALTER TABLE invoice
DROP COLUMN invoice_date;

-- Step 5: Add formatted computed column
ALTER TABLE invoice
ADD formatted_invoice_date AS FORMAT(invoice_date_only, 'dd-MM-yyyy');

-- Verify invoice table
SELECT * FROM invoice;
```

(B) Verifying the cleaned tables

```sql
-- ==============================
-- (E) Verify all 11 tables
-- ==============================
SELECT * FROM album;
SELECT * FROM artist;
SELECT * FROM customer;
SELECT * FROM employee;
SELECT * FROM genre;
SELECT * FROM invoice;
SELECT * FROM invoice_line;
SELECT * FROM media_type;
SELECT * FROM playlist;
SELECT * FROM playlist_track;
SELECT * FROM track;
```

(C) Joining the tables into a single denormalized table called 'MUSIC_DATA'

```sql
-- JOINING THE TABLES TOGETHER TO GET A COMPLETE VIEW OF THE DATA
SELECT
    -- Customer Details
    c.customer_id,
    CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
    c.company,
    c.city AS customer_city,
    c.country AS customer_country,
    c.email,

    -- Employee (Support Rep)
    e.employee_id AS support_rep_id,
    CONCAT(e.first_name, ' ', e.last_name) AS employee_name,
    e.title AS support_rep_title,

    -- Invoice Details
    i.invoice_id,
    i.formatted_invoice_date,
    i.billing_city,
    i.billing_country,
    i.total AS invoice_total,

    -- Invoice Line
    il.invoice_line_id,
    il.unit_price,
    il.quantity,

    -- Track Details
    t.track_id,
    t.name AS track_name,
    t.milliseconds,
    t.unit_price AS track_price,
```

```sql
    -- Album
    a.album_id,
    a.title AS album_title,

    -- Artist
    ar.artist_id,
    ar.name AS artist_name,

    -- Genre
    g.genre_id,
    g.name AS genre_name,

    -- Media Type
    m.media_type_id,
    m.name AS media_type_name,

    -- Playlist
    p.playlist_id,
    p.name AS playlist_name
```

```sql
FROM Customer c
LEFT JOIN Employee e
    ON c.support_rep_id = e.employee_id

LEFT JOIN Invoice i
    ON c.customer_id = i.customer_id

LEFT JOIN Invoice_line il
    ON i.invoice_id = il.invoice_id

LEFT JOIN Track t
    ON il.track_id = t.track_id

LEFT JOIN Album a
    ON t.album_id = a.album_id

LEFT JOIN Artist ar
    ON a.artist_id = ar.artist_id

LEFT JOIN Genre g
    ON t.genre_id = g.genre_id

LEFT JOIN Media_type m
    ON t.media_type_id = m.media_type_id

LEFT JOIN Playlist_Track pt
    ON t.track_id = pt.track_id

LEFT JOIN Playlist p
    ON pt.playlist_id = p.playlist_id
```

```sql
-- Puttng the joined data into a new table called 'MUSIC_DATA' for easier access
SELECT
    c.customer_id,
    CONCAT(c.first_name, ' ', c.last_name) AS customer_name,
    c.company,
    c.city AS customer_city,
    c.country AS customer_country,
    c.email,

    e.employee_id AS support_rep_id,
    CONCAT(e.first_name, ' ', e.last_name) AS employee_name,
    e.title AS support_rep_title,

    i.invoice_id,
    i.formatted_invoice_date,
    i.billing_city,
    i.billing_country,
    i.total AS invoice_total,

    il.invoice_line_id,
    il.unit_price,
    il.quantity,

    t.track_id,
    t.name AS track_name,
    t.milliseconds,
    t.unit_price AS track_price,

    a.album_id,
    a.title AS album_title,

    ar.artist_id,
    ar.name AS artist_name,
```

```sql
    g.genre_id,
    g.name AS genre_name,

    m.media_type_id,
    m.name AS media_type_name,

    p.playlist_id,
    p.name AS playlist_name

INTO MUSIC_DATA

FROM Customer c
LEFT JOIN Employee e ON c.support_rep_id = e.employee_id
LEFT JOIN Invoice i ON c.customer_id = i.customer_id
LEFT JOIN Invoice_line il ON i.invoice_id = il.invoice_id
LEFT JOIN Track t ON il.track_id = t.track_id
LEFT JOIN Album a ON t.album_id = a.album_id
LEFT JOIN Artist ar ON a.artist_id = ar.artist_id
LEFT JOIN Genre g ON t.genre_id = g.genre_id
LEFT JOIN Media_type m ON t.media_type_id = m.media_type_id
LEFT JOIN Playlist_Track pt ON t.track_id = pt.track_id
LEFT JOIN Playlist p ON pt.playlist_id = p.playlist_id
ORDER BY c.customer_id ASC;

SELECT * FROM MUSIC_DATA
ORDER BY customer_id ASC;
```

(D) Business Analysis

Customer Analytics

4.1 Top 10 Customers by Spending
Identifies highest revenue-generating customers.
Business Impact: Helps in loyalty programs and premium targeting strategies.

4.2 Average Customer Lifetime Value (CLV)
Calculates average revenue per customer.
Business Impact: Supports marketing budget allocation and customer acquisition cost decisions.

4.3 Repeat vs One-Time Customers
Classifies customers based on purchase frequency.
Business Impact: Measures customer retention and loyalty levels.

4.4 Revenue per Customer by Country
Identifies countries with high average customer value.
Business Impact: Supports regional investment and expansion strategy.

4.5 Inactive Customers (Last 6 Months)
Finds customers who have stopped purchasing.
Business Impact: Enables re-engagement campaigns and promotional targeting.

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5. Sales & Revenue Analysis

5.1 Monthly Revenue Trends
Analyzes revenue patterns over time.
Business Impact: Identifies seasonality and peak periods.

5.2 Average Order Value (AOV)
Calculates average invoice value.
Business Impact: Helps evaluate pricing strategy and cross-sell effectiveness.

5.3 Revenue by Country & City
Measures geographic revenue contribution.
Business Impact: Supports market prioritization.

5.4 Revenue by Sales Representative
Evaluates employee contribution to revenue.
Business Impact: Supports performance-based incentives.

5.5 Peak Sales Months
Identifies best-performing months or quarters.
Business Impact: Assists in marketing calendar planning.

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6. Product & Content Analysis
6.1 Top Revenue-Generating Tracks
Identifies best-selling tracks.

6.2 Most Purchased Albums
Measures album popularity.

6.3 Tracks Never Purchased
Detects dead inventory.

6.4 Average Track Price by Genre
Analyses pricing strategy across genres.

6.5 Revenue per Track by Genre
Measures genre profitability.
Business Impact: Supports catalogue optimization and pricing decisions.

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7. Artist & Genre Performance
7.1 Top 5 Highest-Grossing Artists
Identifies artists generating most revenue.

7.2 Most Popular Genres by Volume
Measures customer preference.

7.3 Most Popular Genres by Revenue
Identifies high-profit genres.

7.4 Genre Popularity by Country
Combines geography and product demand.
Business Impact: Guides content acquisition and regional marketing.

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8. Employee & Operational Efficiency
8.1 Highest-Spending Customers per Employee
Links employees to high-value clients.

8.2 Average Customers per Employee
Measures workload distribution.

8.3 Revenue by Employee Region
Identifies strong regional performers.
Business Impact: Supports workforce optimization and territory planning.

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9. Geographic Trends
9.1 Customer Distribution by Region
Identifies customer concentration.

9.2 Revenue by Region
Measures market profitability.

9.3 Underserved Regions
Finds areas with many customers but low revenue per customer.
Business Impact: Highlights growth opportunities.

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10. Customer Retention & Purchase Behaviour
10.1 Purchase Frequency Distribution
Measures buying behaviour.

10.2 Average Time Between Purchases
Evaluates customer engagement cycle.

10.3 Multi-Genre Purchase Percentage
Analyses diversity in customer taste.
Business Impact: Supports personalization and recommendation strategy.

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11. Operational Optimization
11.1 Frequently Purchased Track Combinations
Market basket analysis.
Business Impact: Supports recommendation systems.

11.2 Price vs Sales Analysis
Analyzes how pricing affects sales volume.
Business Impact: Supports pricing optimization.

11.3 Media Type Trends Over Time
Tracks format popularity (e.g., MPEG, AAC).
Business Impact: Guides product format strategy.

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12. Key Insights

- Small segment of customers drives major revenue.
- Certain genres and artists dominate profitability.
- Revenue varies significantly by geography.
- Repeat customers are critical to sustained growth.
- Some tracks show zero demand (optimization opportunity).
- Media format trends indicate evolving consumer preferences.